

CLIMATE

ACTION

PASSPORT

AI-powered climate-transition decisions
for Indian MSMEs

AI understands. Data calculates.

Optimization decides. Evidence verifies.

Research • PRD • Architecture • Implementation • Production plan

03 October 2026 • India • Climate Tech • Version 1.0

1. Executive Summary

Climate Action Passport is a decision-support platform for Indian MSMEs that turns messy business data, climate exposure, technology options and financial constraints into a concrete climate-investment plan. The platform is designed around one question: **given this business, its risks, its available budget and its operating constraints, which combination of climate interventions should it implement now, what is the expected financial/environmental effect, and did the result match the prediction?**

The central product differentiator is the **Climate Intervention Portfolio Optimizer**. It is not another carbon calculator, ESG dashboard, generic green score, climate-risk map or finance directory. The optimizer selects feasible combinations of interventions under constraints such as budget, payback, applicability, prerequisites and user priorities.

End-to-end decision loop



NITI Aayog's January 2026 MSME green-transition roadmap explicitly addresses sector-specific technology options, emission baselines, commercial feasibility, capital/funding sources and action plans. BEE's ADEETIE programme combines energy audits, DPRs, financing support and post-implementation monitoring. These sources support the proposition that MSME transition requires implementation-oriented decision infrastructure, not only awareness or measurement. [■cite■turn0search0■turn0search1■](#)

Satin Finserv publicly describes inclusive MSME lending and sustainability financing, including Sustainable & Emerging Businesses and sectors such as EVs, solar and battery energy storage. That makes an MSME climate-transition decision layer a plausible hackathon fit, while the proposed product should remain separate from actual credit approval. [■cite■turn0search4■turn0search5■](#)

Build recommendation: build a focused MVP that demonstrates upload → extraction → baseline → climate risk → optimized portfolio → what-if simulation → passport → evidence → predicted-vs-actual. Do not attempt loan origination, IoT, blockchain, a custom foundation model or a massive climate-data platform in the hackathon.

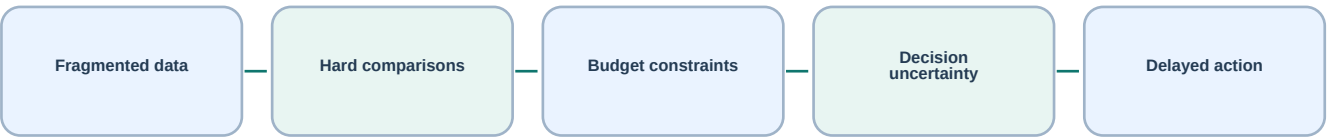
2. Problem Discovery

Problem: MSMEs can face fragmented climate-transition work across bills, resource data, climate-risk information, technology choices, financing documentation and post-project evidence. The practical decision is not only how much carbon is emitted; it is which action is feasible, affordable and useful for this particular business.

Layer	Finding
Root problem	Climate action decisions are fragmented across resource data, risk, technology, finance and evidence.
Symptoms	Manual bill reading; generic recommendations; disconnected risk information; spreadsheet-based comparisons; weak post-project measurement.
Causes	Low data standardization; fragmented tools; limited technical capacity; financing/documentation friction; sector-specific technology differences.
Effects	Slower decisions; uncertain ROI; delayed upgrades; weak traceability of outcomes.

5-Whys: climate action is delayed → decisions are difficult to compare → cost/savings/risk/impact are separated → tools specialize in different layers → MSMEs need an integrated project decision rather than another standalone score.

Problem tree



3. Problem Validation

Verified evidence: NITI Aayog's 2026 roadmap describes the need to identify emission sources, baselines, technology options, commercial feasibility, capital/funding sources and sector-specific action plans. [■cite■turn0search60■](#)

BEE's ADEETIE scheme supports Udyam-registered MSMEs with technical and financial mechanisms including investment-grade energy audits, DPRs, interest subvention and post-implementation monitoring and verification. BEE's technology catalogue includes energy savings, monetary savings, investment and payback fields. [■cite■turn0search1■turn0search11■](#)

The World Bank reported in 2025 that more than 60% of households and firms surveyed across South Asia had experienced extreme weather in the previous five years; this is regional evidence, not an India-only MSME statistic. [■cite■turn0search14■](#)

Dimension	Assessment	Evidence / gap
Problem existence	Strong evidence	Government and programme activity demonstrates a meaningful transition challenge.
User demand	Open	Requires direct interviews or pilot usage; no survey results are invented here.
Technical feasibility	High	Mature web, database, OCR, AI and optimization components exist.
Data feasibility	Moderate	Business documents are obtainable; climate and intervention data need source/version governance.
Commercial feasibility	Promising but unvalidated	Possible B2B/B2B2F channels; willingness-to-pay must be tested.

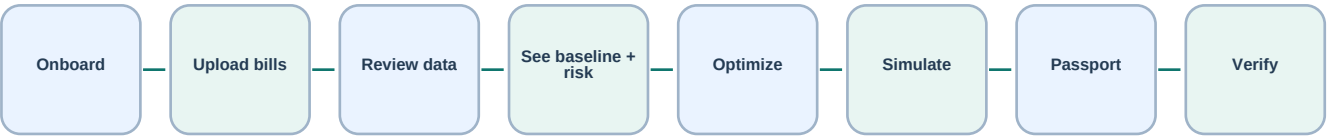
Validation hypothesis: users value a constrained, explainable intervention portfolio more than a conventional climate dashboard or generic recommendation list.

4. User Research

Persona	Goals	Pain points	Decision factors	MVP
Owner/promoter	Lower costs; protect business; invest confidently	Time pressure; fragmented data	Upfront cost; payback; reliability	Primary
Operations manager	Improve energy/resource performance	Manual measurements; equipment uncertainty	Savings; disruption; implementation time	Primary
Accounts/finance	Prepare documents; understand affordability	Bills/invoices scattered	Cash flow; documentation	Secondary
Lender / partner	Understand project and evidence	Inconsistent project files	Risk; documentation; use of funds	Future
Advisor / auditor	Guide projects and verification	Manual assessments	Data quality; methodology	Future

Jobs-to-be-Done: “Help me choose climate upgrades I can afford and operate”; “Help me understand whether the investment will pay back”; “Give me a credible project file I can share with a lender, advisor or partner.”

Primary journey



5. Current-State Analysis

Current step	Typical artifact	Bottleneck	Product opportunity
Collect data	Bills, invoices, spreadsheets	Manual / inconsistent	Document-first ingestion
Assess risk	Maps, reports, advisor input	Separate workflow	Risk layer in project
Find actions	Audits, vendor quotes, lists	Generic / difficult to compare	Intervention KB
Compare economics	Spreadsheets	Scenario changes are manual	Calculation engine + optimizer
Seek finance	DPRs / applications	Documentation friction	Passport
Implement	Invoices / photos / maintenance	Evidence disconnected	Evidence ledger
Measure	Bills / meters	Baseline often missing	Predicted-vs-actual

Future state



6. Competitor Research

Existing solutions show that carbon accounting, climate risk, ESG scoring and green-finance workflows are already active categories. The product must therefore differentiate by decision workflow and optimization rather than by generic AI branding.

Solution / programme	Documented focus	Coverage	Relevant gap
NITI Aayog MSME Roadmap	Policy / transition roadmap	Technology, financing, action planning	Not a business-specific software workflow
BEE ADEETIE	MSME energy efficiency	Audits, DPRs, financing, M&V;	Programme rather than general decision engine
Sustainow Carbon Intelligence	Carbon accounting	Scope 1–3, reporting, finance readiness	Different primary job
Sustainow Klima	Climate risk	Localized physical/transition risk	Risk-centric
Tarulok	MSME ESG	ESG/resource insights and finance	Broader ESG workflow
Vyaparli Pramana	Green creditworthiness	Financial + sustainability score	Scoring, not portfolio optimization
Carbonance Green Finance Suite	Sustainable finance	Eligibility, finance suitability, impact tracking, monitoring	Strong overlap; differentiator is MSME intervention selection + scenario optimization

Carbonance publicly describes project eligibility, sustainable-finance suitability, credit decision support, capital-to-impact tracking and post-financing monitoring; this is why “finance readiness + impact tracking” alone is not sufficient differentiation.

■cite■turn1search0■turn1search5■turn1search10■

Public pricing was not assumed where it was not clearly verified. Competitor feature claims are based on their public product pages, not private demonstrations.

7. Gap Analysis

Gap	Opportunity	Confidence
Decision integration	Connect business data, risk, resource baseline, interventions and finance constraints	High
Budget-constrained selection	Optimize combinations, not just list options	High
Scenario analysis	Re-run plans when budget, tariff or hours change	High
Evidence continuity	Keep project decision and implementation proof together	Medium
Predicted vs actual	Make forecast-to-result loop central	Medium
Micro/small-business simplicity	Progressive disclosure and document-first onboarding	Hypothesis

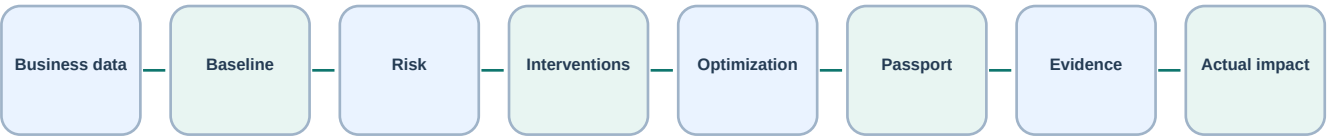
Highest-value opportunity: a traceable, budget-aware intervention decision engine that can be explained in business terms and preserved as a project passport.

8. Proposed Solution

Vision: make climate-transition investment decisions as understandable and evidence-driven for an MSME as a good business budgeting decision.

Module	Purpose
Onboarding	Capture business, operating, resource and financial constraints.
Document intelligence	Extract bills/invoices with confidence and provenance.
Climate risk	Assess heat, flood, water stress and energy vulnerability.
Baseline engine	Calculate resource, cost and emissions baseline.
Intervention KB	Store actions, cost, savings, payback, applicability and evidence.
Portfolio optimizer	Select feasible combinations under explicit constraints.
What-if simulator	Recompute under changed business assumptions.
Climate Action Passport	Versioned project record with impact and evidence.
Evidence + impact	Verify implementation and compare predicted vs actual.

Core value chain



9. Differentiation

Capability	Typical point solution	Climate Action Passport
Carbon accounting	Core	Supporting input
Climate risk	Core	Input to investment decision
Generic recommendations	Common	Converted into feasible portfolios
Budget-aware optimization	Not usually the central job	Core
What-if simulation	Often separate	Core
Project passport	Often report-oriented	Living project object
Evidence after implementation	Varies	Core loop
Predicted vs actual	Varies	Core loop
Credit approval	Some adjacent products	Explicitly outside MVP

Defensibility hypothesis: over time, the system can accumulate structured intervention outcomes by sector, geography and operating profile. The potential moat is therefore the combination of software, intervention ontology, evidence history and outcome data—not any one UI feature.

10. Product Definition

Working name: Climate Action Passport. Alternatives: GreenRoute MSME, ClimateCap, MSME Climate Navigator, TransitionOS.

Product principles: explainable by default; deterministic numbers; AI as assistant; visible assumptions; evidence attached to claims; constraint-aware recommendations; no autonomous credit approval.

Priority	Features
Must Have	Onboarding; documents; extraction; baseline; four risks; intervention KB; optimizer; what-if; passport; evidence; impact.
Should Have	RAG assistant; source explorer; scenario comparison; export; advisor view.
Could Have	Bank/ERP connectors; meters; supplier data; multilingual UX.
Future	IoT; advanced forecasting; financing marketplace; lender APIs; benchmarking.

11. UX & Information Architecture

Navigation: Dashboard → Business → Documents → Risk → Baseline → Interventions → Optimizer → Simulator → Passport → Implementation → Impact.

Screen	Purpose	Core components
Dashboard	Show current state and next action	Risk cards, baseline, active project, pending evidence
Onboarding	Capture minimum viable context	Industry, location, hours, revenue, budget, resource use
Documents	Ingest bills/invoices	Upload, extraction preview, confidence, review
Risk	Explain exposure	Risk cards, drivers, source/date
Baseline	Establish current state	Energy, cost, water, emissions, assumptions
Interventions	Explore actions	Filters, technical details, costs, payback
Optimizer	Choose portfolio	Constraints, top portfolios, rationale
Simulator	Test scenarios	Budget, tariff, hours, priorities
Passport	Create/share project record	Economics, climate impact, milestones, evidence
Implementation	Track delivery	Milestones, evidence, discrepancies
Impact	Compare forecast and reality	Baseline / predicted / actual / variance

UX rule: every decision-relevant number should expose its source, calculation method or assumption. “Why did you recommend this?” is a first-class interaction.

12. Functional Requirements

ID	Requirement	Behavior	Priority
FR-001	Business onboarding	Create/update MSME profile and constraints	Must
FR-002	Document upload	Upload PDF/image bills and invoices	Must
FR-003	Structured extraction	Extract fields with confidence and provenance; no invented values	Must

ID	Requirement	Behavior	Priority
FR-004	Climate risk	Generate four risk dimensions with source/timestamp	Must
FR-005	Baseline	Calculate resource/cost/emissions baseline with versioned factors	Must
FR-006	Intervention generation	Filter applicable actions	Must
FR-007	Portfolio optimization	Return top three feasible portfolios	Must
FR-008	What-if	Recalculate when assumptions change	Must
FR-009	Passport	Create versioned project passport	Must
FR-010	Evidence verification	Compare evidence with plan; flag uncertainty	Must
FR-011	Impact tracking	Compare baseline, predicted and actual	Must
FR-012	AI explanation	Explain outputs without changing numbers	Should
FR-013	Knowledge-base admin	Version interventions and factors	Should
FR-014	Notifications	Remind about missing evidence/measurements	Could

13. Non-Functional Requirements

Area	Requirement	Status
Performance	Interactive API/dashboard calls should generally be low-single-digit seconds excluding AI/OCR jobs	MVP target
Async processing	OCR/extraction/optimization jobs must not block the UI	Required
Availability	Use managed services; formal SLO after pilot traffic is measured	To validate
Security	TLS, auth, RLS, private documents, server-side secrets	Required
Privacy	Data minimization, retention/deletion policy	Required
Accessibility	Keyboard support, labels, contrast, error clarity	Recommended
Auditability	Version factors, interventions, assumptions, passport revisions	Required
Scalability	Stateless API, managed DB, object storage, async jobs	Suitable
Recovery	Automated backups and restore test	Required

14. Technical Architecture

High-level architecture

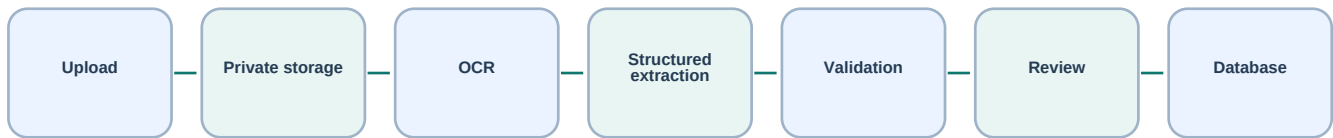


Layer	Recommended technology	Why
Frontend	Next.js + React + TypeScript + Tailwind	Fast iteration and strong component ecosystem
Backend	Python + FastAPI + Pydantic	Strong fit for calculations, data validation and AI orchestration
Database	PostgreSQL / Supabase	Relational integrity for projects, interventions, evidence and impact
Vector retrieval	pgvector	Grounded knowledge retrieval without another database
Storage	Supabase Storage / S3	Private documents and evidence
AI	OpenAI API	Extraction, explanation and RAG
OCR	Document AI / Textract / Azure DI / Tesseract	Text/layout extraction
Optimization	Google OR-Tools	Constraint optimization
Maps	Leaflet + OSM or Mapbox	Risk/location visualization
Deployment	Vercel + managed backend + Supabase	Low operational burden
Observability	Sentry + structured logs	Error and operational visibility
CI/CD	GitHub Actions	Repeatable build/test/deploy

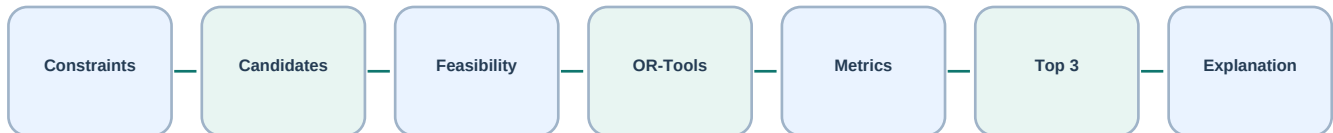
Architecture principle: LLM ≠ decision engine. LLMs handle document understanding, retrieval and explanation. Deterministic Python calculates economics/environmental metrics. OR-Tools selects portfolios.

15. Data Flow & Sequence Design

Document intelligence



Optimizer



Evidence



Document sequence: store the original → OCR → strict JSON extraction → schema validation → field-level provenance/confidence → human review for uncertainty → calculation engine.

Optimization sequence: load current state → filter candidate interventions → solve explicit objective under constraints → calculate projected metrics → return portfolios and explanation metadata.

16. Database Design

Entity	Key fields	Relationship
users	id, auth_id, role	Owns businesses/projects
businesses	id, name, industry, location, hours, budget	Has documents, risks, projects
documents	id, business_id, type, storage_uri, status	Has extractions
extractions	document_id, field, value, confidence, provenance	Belongs to document
resource_records	business_id, period, electricity, water, fuel, cost	Feeds baseline
climate_risks	business_id, type, score, drivers, source, timestamp	Belongs to business
interventions	id, name, applicability, cost, payback, parameters	Versioned KB
projects	id, business_id, budget, status	Owns passport/evidence/impact
project_interventions	project_id, intervention_id, quantity, cost	Many-to-many
evidence	project_id, type, source_uri, extracted values, confidence	Supports implementation
impact_records	project_id, period, baseline, predicted, actual	Measures outcome
factor_versions	type, value, unit, source, effective_date	Supports calculations
audit_logs	actor, action, entity, timestamp	Cross-cutting

SQL vs NoSQL: PostgreSQL is preferred because the core domain is relational and auditability/transaction integrity matter. JSONB can hold semi-structured extraction payloads without abandoning relational constraints.

17. API Design

Method	Endpoint	Purpose
POST	/api/v1/businesses	Create business
GET	/api/v1/businesses/{id}	Get business
POST	/api/v1/documents	Upload / process document
GET	/api/v1/documents/{id}	Get extraction state

Method	Endpoint	Purpose
POST	/api/v1/documents/{id}/review	Approve/edit extraction
GET	/api/v1/risks/{business_id}	Get climate risks
GET	/api/v1/baseline/{business_id}	Get baseline
GET	/api/v1/interventions	Search intervention KB
POST	/api/v1/optimizer/run	Generate portfolios
POST	/api/v1/simulations	Run what-if scenario
POST	/api/v1/projects	Create project/passport
POST	/api/v1/projects/{id}/evidence	Add evidence
GET	/api/v1/projects/{id}/impact	Get impact

API conventions: /api/v1 versioning; pagination for lists; idempotency keys for processing and creation; structured errors; authorization on every tenant-scoped object; rate limiting; no client-side authorization decisions.

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "available_budget must be greater than 0",
    "fields": {"available_budget": "positive number required"},
    "request_id": "req_123"
  }
}
```

18. AI / ML Strategy

Use case	AI?	Implementation	Guardrail
Bill extraction	Yes	OCR + strict structured LLM extraction	Schema validation + human review
Risk explanation	Optional	RAG over sourced data	Never invent source/score
Intervention explanation	Yes	RAG over intervention KB	Numbers from calculation engine
Portfolio optimization	No	OR-Tools	Explicit constraints/objective
ROI/emissions math	No	Python engine	Versioned factors + tests
Forecasting	Future	ML after enough data	Do not add to MVP
Causal attribution	Future	Statistical/causal methods	Do not infer from simple before/after

AI evaluation: field-level extraction accuracy, missing-field detection, confidence calibration, grounding rate, unsupported-claim rate and numerical consistency. A versioned benchmark set should include representative bills and deliberately difficult documents.

Prompt injection: uploaded documents are untrusted content. Never execute instructions found inside a bill or invoice. Use strict schemas, isolate document content from system instructions, limit tool access and validate all outputs.

19. Security & Privacy

Threat	Example	Control
Spoofing	Unauthorized login	Managed auth; secure sessions; optional MFA
Tampering	Changed project numbers	RLS; audit logs; versioned calculations
Repudiation	Disputed edits	Timestamped audit trail
Disclosure	Private bills exposed	Private storage; signed access; least privilege
Denial of service	Upload/API abuse	Rate limits; file-size limits; quotas
Elevation	Cross-business access	Tenant isolation; RLS; authorization checks
Prompt injection	Malicious document text	Untrusted-content boundary; strict schemas
Supply chain	Compromised dependency	Lockfiles; dependency scanning; CI

Financial boundary: do not claim loan approval, guaranteed scheme eligibility or creditworthiness from a climate score. The product creates project decision support and evidence infrastructure; regulated credit decisions remain with the relevant institution.

20. Testing Strategy

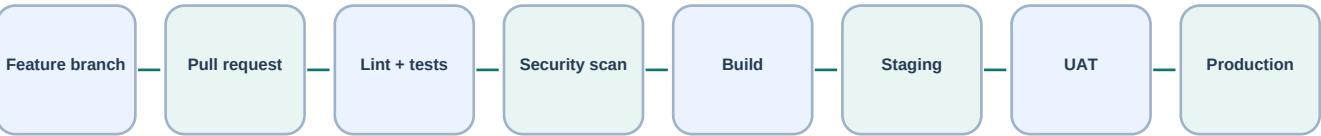
Layer	Tests	Quality gate
Unit	Payback, ROI, emissions, water, risk normalization	Deterministic tests pass
Integration	Upload → OCR → extraction → DB → baseline	Data integrity
API	Auth, validation, errors, rate limits	Contract tests

Layer	Tests	Quality gate
E2E	Onboard → upload → optimize → passport	Critical path
AI	Extraction, missing fields, grounding, hallucination	Benchmark threshold
Security	Auth, file, injection, access control	No critical findings
Accessibility	Keyboard, labels, contrast, errors	Core screens usable
Performance	API/load tests	Measured against SLO
UAT	Owner completes core journey	Acceptance criteria

Critical acceptance test: a seeded fictional textile MSME can upload three synthetic bills, review extracted values, obtain a baseline, see risks, generate portfolios, change the budget, see a changed scenario, generate a passport and mark implementation evidence. All numbers must originate from the same backend calculation versions.

21. DevOps & Production Deployment

CI/CD



Recommended approach: short-lived branches, protected main, automated tests before merge, staging environment, managed secrets and backward-compatible database migrations.

Deployment: Vercel for frontend; managed FastAPI hosting such as Render/Railway/AWS; Supabase/Postgres; private object storage. Rolling deployment is sufficient for the MVP. Blue-green or canary deployment can be introduced when traffic and operational risk justify it.

Rollback: application rollback to a known build; preserve historical passport and factor versions; use reversible/backward-compatible migrations wherever practical.

22. Monitoring & Observability

Layer	Metrics
Application	Request volume, latency, errors, queue age
AI/OCR	Processing time, failure rate, low-confidence rate, model/token cost
Optimizer	Solve time, infeasible requests, constraint violations
Database	Latency, connections, storage, backup health
Business	Onboarding completion, documents approved, optimizer runs, passports, evidence completion
Impact	Predicted-vs-actual variance, measurement coverage

Alerts: API error spikes, processing backlog, database failure, abnormal authorization failures, AI extraction degradation, optimizer infeasibility spikes and storage anomalies.

23. Cost Analysis

These are planning ranges, not vendor quotations. Actual cost depends on traffic, document volume, model/OCR provider, storage and retention.

Stage	Cost shape	Drivers
Hackathon	Near-zero to low monthly	Free tiers; limited AI/OCR; synthetic data
Pilot	Potentially low hundreds USD/month	AI/OCR, storage, DB, monitoring
Small production	Potentially hundreds to low thousands USD/month	Inference volume, redundancy, support
Scale	Highly variable	Documents, integrations, inference, retention, support

Cost controls: batch extraction; use smaller models when sufficient; cache deterministic results; avoid repeated document transmission; asynchronous jobs; usage budgets; cost-per-business and cost-per-optimizer-run dashboards.

24. Business Model

Canvas element	Proposal
Customer segments	Micro/small MSMEs; lenders; energy/climate advisors; cluster organizations
Value proposition	Convert fragmented climate/resource data into feasible investment plans with evidence
Channels	Lenders; MSME associations; advisors; cluster programmes; self-serve
Relationships	Guided onboarding; advisor-assisted workflows; recurring impact tracking
Revenue hypotheses	SaaS; B2B licensing; per-project fee; partner/API fee
Key resources	Software; intervention KB; factor registry; climate adapters; outcome data
Partners	Energy auditors; technology vendors; lenders; climate-data providers; MSME ecosystems
Costs	Cloud; AI/OCR; engineering; support; data; compliance

Commercial hypothesis: B2B2F may be attractive because a structured project passport can sit near financing and advisory workflows. Direct MSME SaaS is another hypothesis. Neither should be treated as validated until interviews/pilots establish willingness-to-pay and acquisition economics.

25. Impact Analysis

KPI	Definition	Source
Energy savings	Baseline energy – measured post-intervention energy	Bills/meters
Energy cost savings	Baseline cost – actual cost, with tariff assumptions	Bills/tariff data
CO2 avoided	Baseline emissions – post-intervention emissions using factor version	Resource data + factor registry
Water savings	Baseline water – actual water	Bills/meters
Investment delivered	Verified spend	Invoices/evidence
Prediction accuracy	Actual – predicted / predicted	Impact records
Evidence completeness	Verified evidence / required evidence	Evidence ledger

Measurement discipline: distinguish predicted, measured and attributed outcomes. A reduction before/after an intervention is not automatically causal proof. Stronger attribution can be added later with normalization, controls or causal methods.

26. MVP Definition

MVP success criterion: a judge or pilot user can provide a business profile and bills, understand the baseline and risks, see a constrained intervention portfolio, change a scenario, and receive a traceable project passport without the team manually changing numbers behind the scenes.

Include	Exclude
Onboarding + document upload	Real loan approval
OCR/AI extraction + review	Live bank account integrations
Four climate risks	IoT streaming
20–30 seeded interventions	Huge intervention marketplace
Deterministic calculation engine	Complex forecasting ML
OR-Tools optimizer	Autonomous procurement
What-if simulator	Blockchain
Passport + evidence + impact	Full regulatory certification

Use a clearly labelled fictional MSME and synthetic data for the hackathon demo. Synthetic values must never be presented as real-world measurements.

27. Implementation Roadmap

Phase	Focus	Tasks	Exit
0	Scope + evidence	Freeze MVP; factors; sources; demo data	Approved PRD
1	UX	Flows; wireframes; design system	Clickable core journey
2	Foundation	Repo; auth; DB; storage; CI	Deployable skeleton
3	Ingestion	Upload; OCR; extraction; review	Structured bills
4	Climate + baseline	Risk adapters; calculations	Trusted baseline
5	Intervention KB	Seed + version actions	Candidate set
6	Optimizer	Constraints; objective; OR-Tools	Top 3 portfolios

Phase	Focus	Tasks	Exit
7	Simulator + passport	Scenarios; project record; export	End-to-end demo
8	Evidence + impact	Verification; actuals	Closed loop
9	Hardening	Tests; security; monitoring	Pilot-ready

28. Risk Analysis

Risk	Probability	Impact	Mitigation
Bad extraction	Med	High	Schema validation; confidence; review
Weak data sources	Med	High	Source registry; timestamps; fallback adapters
AI overreach	Med	High	LLM never calculates core metrics
Odd optimizer output	Med	High	Constraint tests; explicit objective; fallback heuristics
User distrust	Med	High	Show assumptions; evidence; advisor review
Scope creep	High	High	Strict MVP exclusions
Security breach	Low/Med	Very High	RLS; private storage; scanning; least privilege
Cost escalation	Med	Med	Usage budgets; monitoring; model fallback
False climate claim	Med	High	Source/version registry; mark unknown

29. Production Readiness Checklist

Area	Checklist	Status
Product	Core user journey end-to-end	Not Started
UX	Empty/loading/error/success states	Not Started
Backend	Validation + authorization	Not Started
Frontend	Responsive/accessibility core screens	Not Started
Database	Migrations; indexes; backups	Not Started
API	Versioning; idempotency; rate limits	Not Started
Security	Threat model; secrets; RLS	Not Started
Testing	Unit/integration/E2E critical path	Not Started
Infrastructure	Staging + production	Not Started
CI/CD	Automated tests/build	Not Started
Monitoring	Errors; latency; AI/OCR jobs	Not Started
Backup	Restore test	Not Started
DR	RPO/RTO documented	Not Started
Documentation	README; architecture; API; data dictionary	Not Started
Legal/compliance	Privacy/retention review	Not Started
Support	Incident/contact process	Not Started
Analytics	Product + impact KPIs	Not Started

30. Final Feasibility Assessment

Dimension	Assessment	Reason
Problem severity	8/10	Strong ecosystem evidence; exact user pain intensity still needs direct interviews.
Market opportunity	7/10	Large MSME ecosystem and active transition/finance initiatives; willingness-to-pay unvalidated.
User demand	6/10	Plausible but not directly validated in this project.
Technical feasibility	9/10	Mature components support the core workflow.
Business feasibility	7/10	Several plausible channels; partner economics need validation.
Scalability	8/10	Stateless APIs + managed relational data + async jobs are scalable.
Innovation	8/10	Workflow and constrained portfolio decision support provide the main innovation.
Differentiation	8/10	Clear wedge if optimizer + simulator + passport + outcome loop are executed well.
Implementation difficulty	7/10	Data quality, climate methodology and AI reliability are non-trivial.
Production readiness	5/10	Architecture is production-oriented but still requires hardening and pilot validation.

Dimension	Assessment	Reason
Overall potential	8/10	Strong hackathon concept with a credible pilot path if scope remains focused.

Build decision: MODIFY → BUILD. Build the focused decision-support MVP rather than the full climate/finance platform. The key hypothesis is that users value a feasible, budget-aware intervention portfolio more than a conventional dashboard or generic recommendation list.

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Carbonance — Corporate Carbon Accounting Solutions.
<https://carbonance.com/en/ccas/>

Uploaded project brief — requested 30-phase analysis and final 32-section structure.
Conversation attachment: Pasted text.txt

32. Appendices

A. Mermaid-compatible architecture diagrams

```
flowchart LR
  U[MSME User] --> FE[Next.js / React]
  FE --> API[FastAPI]
  API --> DB[(PostgreSQL / Supabase)]
  API --> ST[(Private Object Storage)]
  API --> OCR[OCR / Document AI]
  API --> LLM[LLM Extraction + RAG]
  API --> OPT[OR-Tools Optimizer]
  API --> CL[Climate Data Adapters]
  API --> CALC[Deterministic Calculation Engine]
  OPT --> CALC
  CALC --> DB
```

B. Optimizer diagram

```
flowchart TD
  A[Business Profile] --> B[Constraints]
  C[Intervention Knowledge Base] --> D[Candidate Filter]
  B --> D
  D --> E[OR-Tools Optimization]
  E --> F[Top 3 Portfolios]
  F --> G[Deterministic Metrics]
  G --> H[Explanation + Passport]
```

C. ER diagram

```
erDiagram
    BUSINESS ||--o{ DOCUMENT : has
    BUSINESS ||--o{ CLIMATE_RISK : has
    BUSINESS ||--o{ PROJECT : owns
    PROJECT }o--o{ INTERVENTION : selects
    PROJECT ||--o{ EVIDENCE : requires
    PROJECT ||--o{ IMPACT_RECORD : measures
    DOCUMENT ||--o{ EXTRACTION : yields
    FACTOR_VERSION ||--o{ IMPACT_RECORD : supports
```

D. Intervention knowledge-base schema

```
{
  "id": "INT-001",
  "name": "High-efficiency motor upgrade",
  "category": "energy_efficiency",
  "applicable_industries": ["textile", "manufacturing"],
  "investment_range": [50000, 500000],
  "payback_years": 2.8,
  "energy_saving_pct": 8,
  "water_saving_pct": 0,
  "prerequisites": ["motor_inventory"],
  "climate_benefits": ["energy_vulnerability"],
  "source": "technical source / audit / vendor",
  "source_year": 2026,
  "confidence": "medium",
  "demo_assumption": true
}
```

E. Optimizer objective

```
maximize(
  w_financial * normalized_annual_savings
+ w_carbon * normalized_avoided_co2
+ w_water * normalized_water_savings
+ w_resilience * normalized_risk_reduction
)
```

```
subject to:
investment <= available_budget
payback <= max_payback
applicability == true
prerequisites satisfied
incompatibilities excluded
```

Weights and factor versions are stored with each scenario.

F. Recommended implementation prompts

Prompt 1

Build the Next.js + TypeScript frontend for dashboard, onboarding, documents, risk, baseline, interventions, optimizer, simulator, passport, implementation and impact. Use mock data first, reusable components and explicit API boundaries. Never calculate core business metrics in the frontend.

Prompt 2

Implement document ingestion: private upload → OCR adapter → strict structured extraction → Pydantic validation → field-level confidence/provenance → review UI → database. Never invent missing values.

Prompt 3

Implement deterministic calculations for energy cost, baseline emissions, projected energy/emissions, avoided emissions, water savings, annual savings, payback and ROI. Version factors and assumptions. Add unit tests.

Prompt 4

Seed 20–30 intervention records with applicability, investment range, payback, energy/water/CO2 parameters, prerequisites, incompatibilities, climate benefits, source/year and confidence. Mark demo assumptions.

Prompt 5

Implement the Climate Intervention Portfolio Optimizer using OR-Tools. Inputs: constraints, interventions, budget, max payback and objective weights. Return top three feasible portfolios with metrics, constraints, assumptions and explanation. Do not use an LLM to choose.

Prompt 6

Implement the what-if simulator. Inputs: budget, electricity price, operating hours, max payback and priority weights. Re-run backend optimization and compare scenarios.

Prompt 7

Implement the Climate Action Passport as a versioned project object with business context, risks, baseline, selected portfolio, investment, predicted impact, payback, milestones, evidence requirements, sources and assumptions. Never claim loan approval.

Prompt 8

Implement evidence verification and predicted-vs-actual tracking. Label uncertain evidence Needs Review. Preserve measurement period and source. Do not imply causality beyond the data.

Prompt 9

Implement security: managed auth, RLS, private storage, server-side secrets, validation, rate limits, audit logs, secure file handling and dependency scanning. Treat uploaded text as untrusted prompt-injection content.

Prompt 10

Create CI/CD with lint, unit tests, integration tests, build checks and dependency/security scans. Add staging deployment, environment-specific secrets and rollback documentation.

Final self-review: the document distinguishes verified evidence from hypotheses and planning estimates; avoids claiming complete uniqueness; keeps credit approval outside the product boundary; separates LLM tasks from deterministic calculations; and follows the requested research → product → architecture → production sequence.